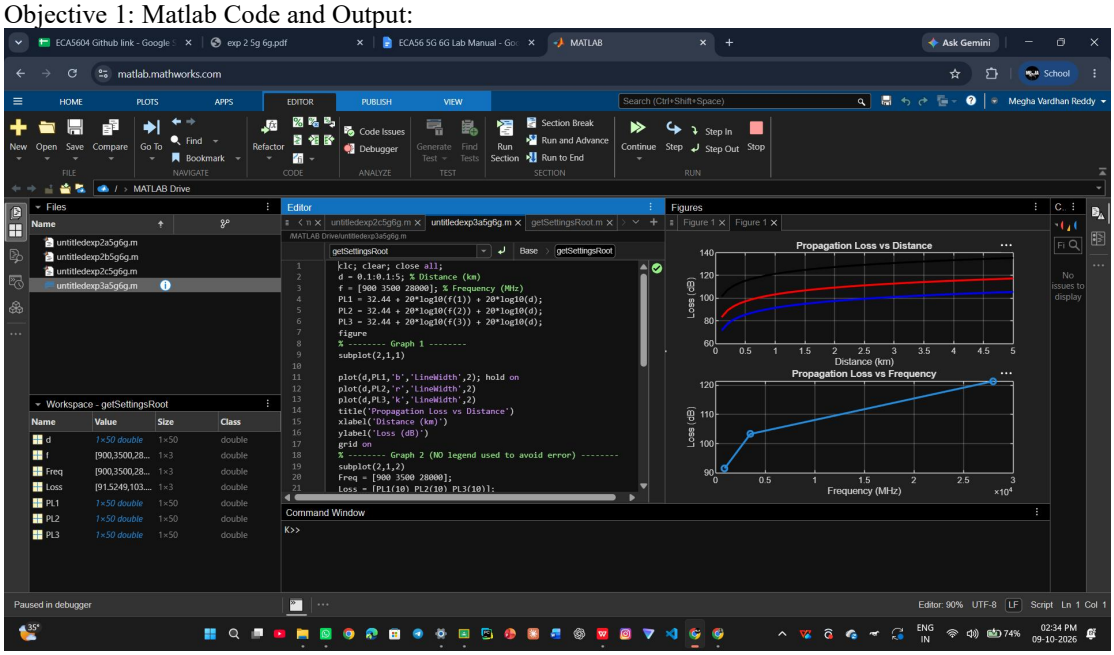
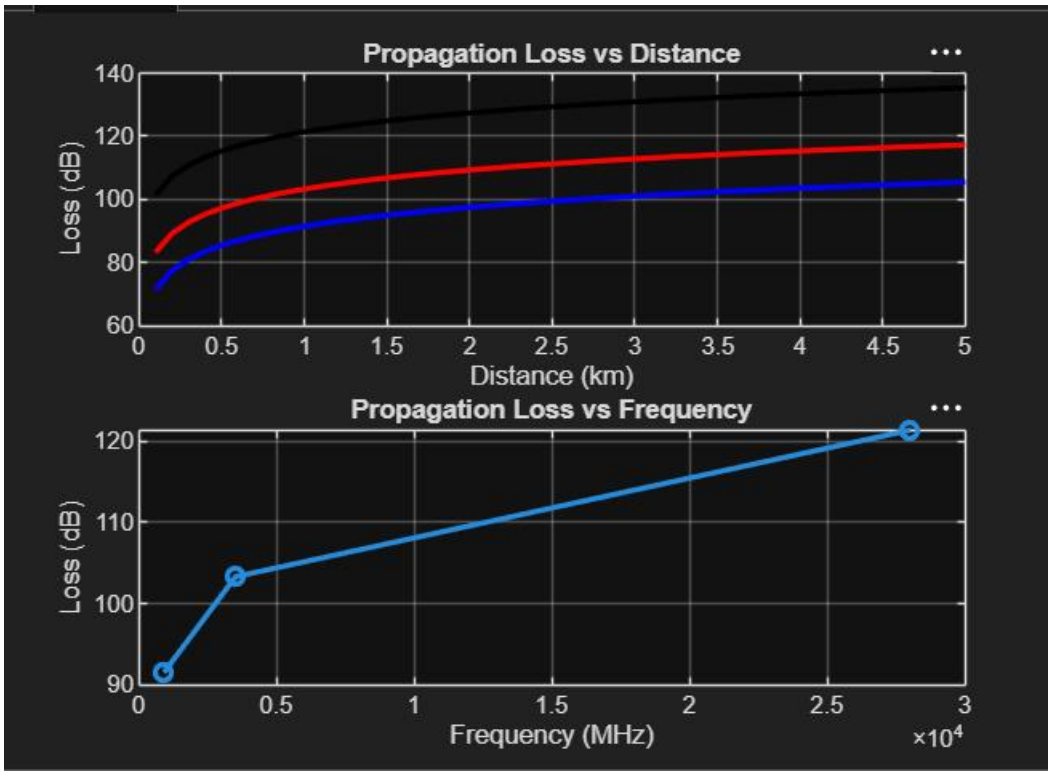
Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss.

Objective :

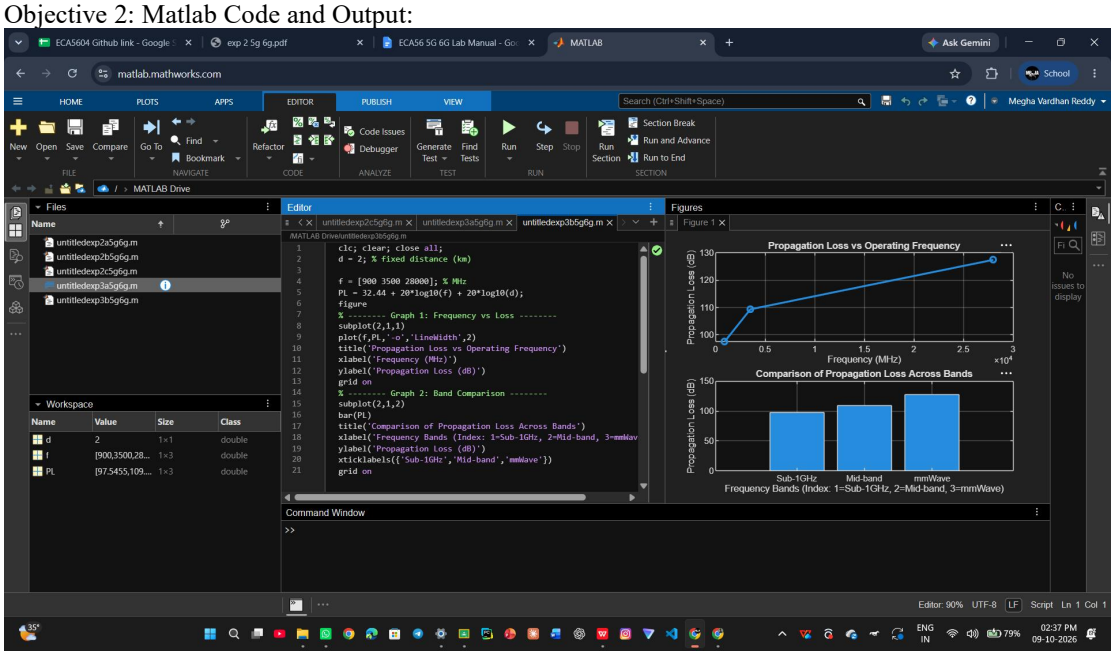
1. To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.
2. To evaluate the impact of different Operating Frequencies (GHz) on signal attenuation and coverage performance.
3. To analyze how Communication Distance (km) affects propagation loss across different spectrum bands.

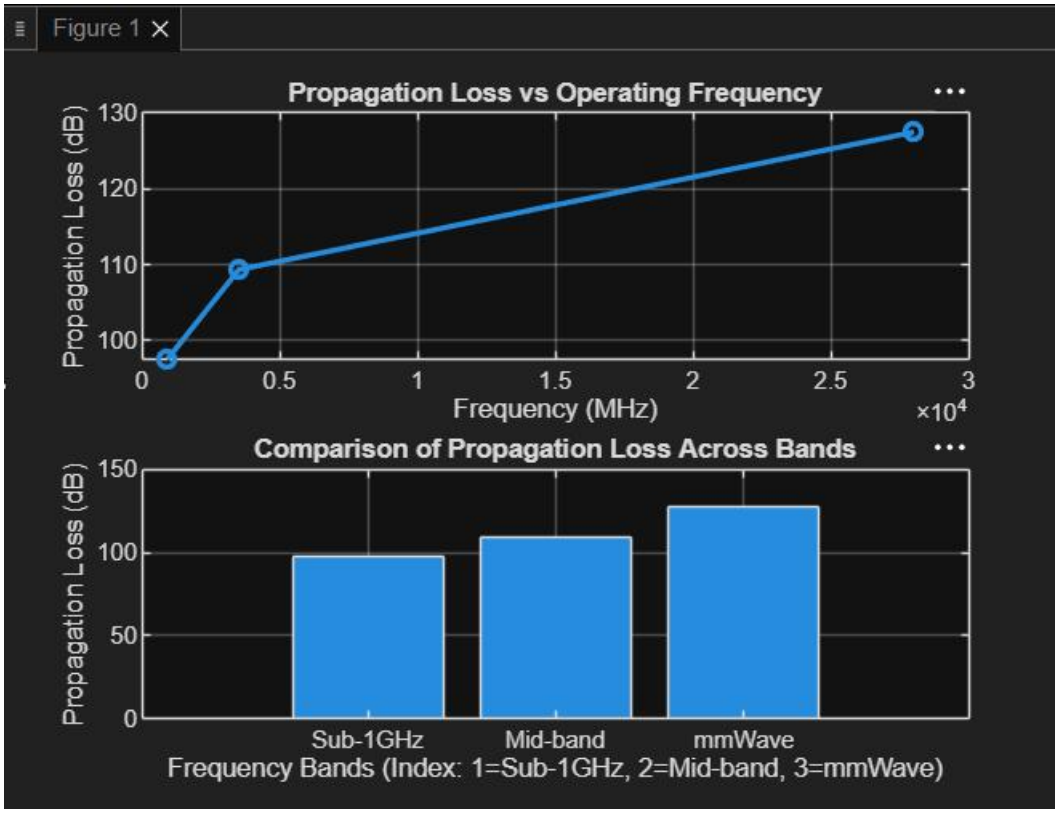
Objective 1: Matlab Code and Output:



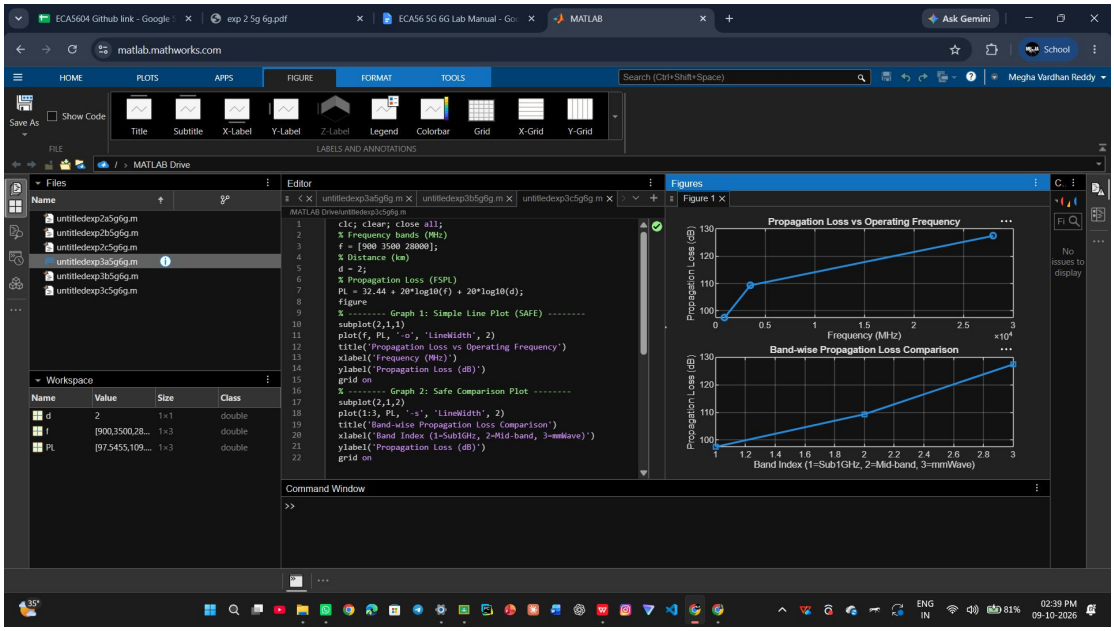


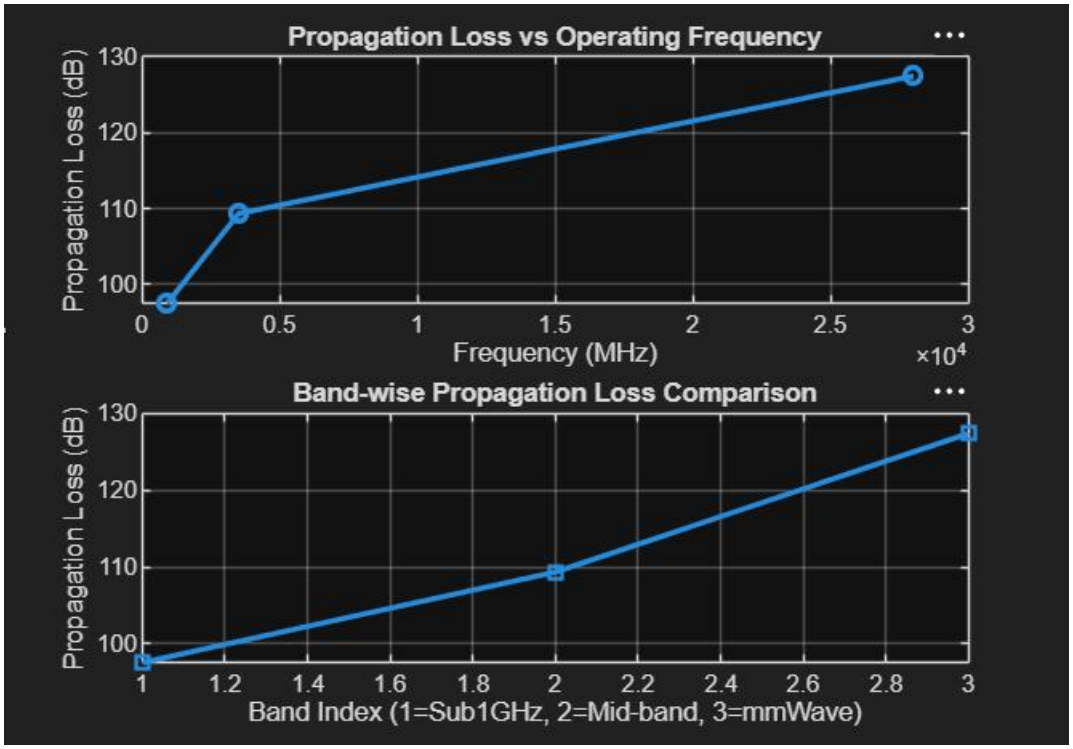
Objective 2: Matlab Code and Output:





Objective 3: Matlab Code and Output:


ECA5604 Github link - Google
exp 2 5g 6g.pdf
ECA56 5G 6G Lab Manual - Goo
MATLAB
Ask Gemini
matlab.mathworks.com
School
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Megha Vardhan Reddy
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
untitledexp2a5g6g.m
untitledexp2b5g6g.m
untitledexp2c5g6g.m
untitledexp3a5g6g.m
untitledexp3b5g6g.m
untitledexp3c5g6g.m
Workspace
Name
Value
Size
Class
d
2
1×1
double
f
[900,3500,28...
1×3
double
PL
[97.5455,109....
1×3
double
Editor
untitledexp3a5g6g.m
untitledexp3b5g6g.m
untitledexp3c5g6g.m
clc; clear; close all;
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Simple Line Plot (SAFE) --------
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
Figures
Figure 1
Propagation Loss vs Operating Frequency
Propagation Loss (dB)
Frequency (MHz)
Band-wise Propagation Loss Comparison
Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)
Command Window
>>
ENG
IN
81%
02:39 PM
09-10-2026


Propagation Loss vs Operating Frequency
Propagation Loss (dB)
Frequency (MHz)
×10^4
Band-wise Propagation Loss Comparison
Propagation Loss (dB)
Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)